

Lecture 5

Openness

- Recall that the r -neighborhood (abbreviated nbh) of $p \in M$ is $\{x \in M : d(x, p) < r\}$
- An **open set** in M is a subset $U \subset M$ s.t. $\forall p \in U \exists r > 0 (M_{r(p)} \subset U)$
 - Think of an area as a metric space. Draw some region inside. It is open if all points inside have some r -neighborhood that all points of M inside that disc are also in U .
 - For $\mathbb{R}$, the open interval (a, b) is an open set
 - $\forall p \in (a, b)$, we can choose some r that $(p - r, p + r) \subset (a, b)$
 - $\{r \in \mathbb{Q} : a < r < b\}$
 - In the metric space of $\mathbb{Q}$, this set is open
 - In the metric space of $\mathbb{R}$, this set is closed. (All irrationals are not included, so no r -neighborhoods work since they all contain irrationals)

Cauchy Sequences

- A sequence (x_n) in metric space M that satisfies the condition: $\forall \varepsilon > 0 \exists N \in \mathbb{N}$ s.t. $\forall m, n \geq N$ we have $d(x_n, x_m) < \varepsilon$
- If all Cauchy Sequences in M converge to limits in M then M is complete with respect to Cauchy Sequences
- $\mathbb{Q}$ is not complete
 - Can show completeness using Cauchy sequences
 - Must mesh different sequences, each real number is an equivalence class of Cauchy Sequences
 - Must easier to show field, LUB properties of $\mathbb{R}$
- $\mathbb{R}$ is complete with respect to Cauchy sequences
 - Can use Peaks
 - Every sequence of real numbers has a monotone subsequence
 - $[a, b]$ is compact
 - Any sequence in $[a, b]$ has a monotone subsequence and must be bounded, and all bounded monotone sequences in $\mathbb{R}$ converge
 - Every Cauchy Sequence in $\mathbb{R}$ converges to a limit in $\mathbb{R}$
 - Every Cauchy sequence in $\mathbb{R}$ is bounded (prove this)
 - Has monotone subsequence by previous idea, terms tend to same limit
 - Hand-wavy way to think about it is that both the sequence and a bounded monotone sequence get close to some value
 - $\mathbb{R}$ has LUB property but $\mathbb{Q}$ doesn't

Theorem: For every metric space M , $M_{r(p)}$ is an open set in M (“Something that would be easy to put on the midterm”)

- Think of a r -nbh in $2d$, would be a disc
- Take some point q in this r -nbh, we can use triangle inequality
- $d(p, q) < r$ by definition, so we can take some value $s = r - d(p, q)$
- $M_{s(q)}$ is entirely contained in $M_{r(p)}$

Topology

- $\mathcal{U}$ is the collection of all open sets of M
- Fulfills 3 conditions:
 - $\emptyset, M \in \mathcal{U}$
 - Every union of open sets is open
 - Any finite intersection of open sets is open

- A topological space is defined by the 3 conditions listed above
- $\mathcal{U}$ is called the topology of M
- Interesting to look at : Urysohn - something about topological spaces being metrizable
- Can show why M is a topological space
 - Condition 1: obvious
 - Condition 2: can break down into original sets when analyzing each specific point
 - Condition 3: for each point take the smallest r -nbh of all the intervals **does this necessarily work? Can it be that the smallest r -nbh is outside the intersection?**
 - Cannot be infinite. Intersections of all $(-\frac{1}{n}, \frac{1}{n})$ for natural numbers n is a point, which is closed.

Closedness

- $K \subset M$ is closed (in M) iff its complement in M is open
- A set can be both open and closed or neither
- Every metric space contains subspaces that are both open and closed
- Since $\emptyset$ is open and $M = \emptyset^C$, M is closed. M is also open. Thus, M is **clopen**.

Limit Points

- $X \subset M, p \in M$ is a limit point of X iff $\exists$ sequence (x_n) in X such that $\lim_{n \rightarrow \infty} x_n = p$
 - Some books require no repetition in this sequence, but not Pugh
- p is not necessarily in X
- $\lim(X) = \{\text{limit points of } X\}$
- Condensation point
 - “Super” limit point
 - Set S “condenses” to that point
 - Uncountable # of points of X in every nbh
- Theorem: X is closed in M iff $X = \lim(X)$